Magnesium sulfate: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Magnesium sulfate: Patient drug information" and "Magnesium sulfate: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Epsom Salt [OTC];
- FT Epsom Salt [OTC];
- GoodSense Epsom Salt [OTC];
- M2 Magnesium [OTC] [DSC]

Pharmacologic Category

- Antiseizure Agent, Miscellaneous;
- Electrolyte Supplement, Parenteral;
- Magnesium Salt

Dosing: Adult

Dosage guidance:

Dosing: All doses in this monograph are expressed as magnesium sulfate salt unless stated otherwise. One gram of magnesium sulfate salt = 98.6 mg of elemental magnesium = 8.12 mEq of elemental magnesium = 4.06 mmol of elemental magnesium.

Clinical considerations: Serum magnesium levels do not correlate well with total body stores as the majority of magnesium is intracellular; serum concentrations may be transiently elevated for a few hours after administration of an IV dose (Ref).

Asthma, severe acute exacerbations

Asthma, severe acute exacerbations (adjunctive agent) (off-label use):

Note: For severe or life-threatening exacerbations nonresponsive to initial therapy. Not recommended for routine use (Ref).

IV: 2 g as a single dose over 20 minutes (Ref).

Chronic obstructive pulmonary disease, severe acute exacerbation

Chronic obstructive pulmonary disease, severe acute exacerbation (adjunctive agent) (off-label use):

Note: For severe or life-threatening exacerbations nonresponsive to initial therapy. Not recommended for routine use (Ref).

IV: 2 g as a single dose over 20 minutes (Ref).

Constipation

Constipation (alternative agent):

Note: For occasional use only. Use in some patients (eg, in those with cardiac dysfunction, kidney disease) may result in electrolyte disturbances and volume overload (Ref).

OTC labeling (patient-guided therapy): **Oral:** 2 to 4 level teaspoons (~10 to 20 g) of granules dissolved in 8 ounces (240 mL) of water; may repeat in 4 hours. Do not exceed 2 doses per day (Ref).

Eclampsia/Preeclampsia with severe features, seizure prophylaxis and treatment

Eclampsia/Preeclampsia with severe features, seizure prophylaxis and treatment:

Note: An optimal regimen has not been identified. Close monitoring (including kidney function [eg, urine output], respiration, and patellar reflexes) is required; monitoring of magnesium levels is recommended for patients with kidney impairment and in patients with signs of toxicity or seizure(s). Adjust dose to avoid maternal toxicity. Calcium gluconate should be available for treating severe magnesium toxicity (Ref).

IV: Initial: 4 to 6 g loading dose over 15 to 30 minutes at onset of labor or induction/cesarean delivery, followed by 1 to 2 g/hour continuous infusion for at least 24 hours after delivery. If seizure occurs while receiving magnesium, an additional bolus of 2 to 4 g may be administered over ≥5 minutes, and continuous infusion may be increased up to a maximum rate of 3 g/hour, with frequent monitoring for toxicity (Ref). Product labeling recommends a maximum dose of 40 g per 24 hours; however, this varies among regimens and institutional protocols.

IM (50% concentration): Initial: 10 g loading dose administered as 5 g in each buttock at onset of labor or induction/cesarean delivery, followed by 5 g every 4 hours for at least 24 hours after delivery. **Note:** Use IM route when unable to establish venous access (Ref).

Fetal neuroprotection for imminent preterm birth

Fetal neuroprotection for imminent preterm birth (maternal administration) (off-label use):

Note: An optimal regimen has not been identified. Close monitoring (including kidney function [eg, urine output], respiration, and patellar reflexes) is required (Ref). Generally used for pregnancies <32 weeks' gestation with expected delivery within 24 hours (Ref). Regimens vary; consult institutional protocols. An example regimen is listed below:

IV: Initial: 4 g loading dose over 20 to 30 minutes (Ref); may follow with a 1 g/hour continuous infusion for a maximum of 24 hours or until delivery, whichever comes first (Ref).

Hypomagnesemia, treatment

Hypomagnesemia, treatment:

Note: Recommended dosing and infusion rates may vary among institutional protocols according to severity and clinical status (Ref). Continuous cardiac monitoring is strongly recommended in symptomatic patients. Subsequent dosing may be based on serum magnesium levels assessed at least 6 to 12 hours after initial dosing. Repletion may take several days (Ref).

Hemodynamically stable patients:

Note: Oral replacement using a different salt (eg, magnesium chloride) is generally preferred for patients with no or minimal symptoms if tolerated (Ref). Use IM only if IV access is unavailable (Ref).

Initial:

Mild deficiency (eg, serum magnesium 1.5 to 1.7 mg/dL):

IV: 1 to 2 g over 1 to 2 hours (Ref).

IM (50% concentration): 1 g every 6 hours for 4 doses (Ref).

Moderate deficiency (eg, serum magnesium 1 to 1.4 mg/dL):

IV: 2 to 4 g over 2 to 12 hours (Ref).

IM (50% concentration): 1 g every 6 hours for 4 doses (Ref).

Severe deficiency (eg, serum magnesium <1 mg/dL):

IV: 4 to 8 g over 4 to 24 hours (Ref). For symptomatic patients, some experts recommend 1 to 2 g over 5 to 60 minutes, followed by an additional 4 to 8 g over 12 to 24 hours (Ref).

Hemodynamically unstable patients:

IV: 1 to 2 g administered as a bolus over 2 to 15 minutes; may repeat as needed if patient remains unstable; once patient is stable, administer an additional 4 to 8 g over 12 to 24 hours (Ref).

Parenteral nutrition

Parenteral nutrition:

Note: Parenteral nutrition ordering requires advanced knowledge of nutrient and other metabolic requirements and should only be prescribed by clinicians trained in assessment and order writing for parenteral nutrition (Ref).

*Standard daily magnesium requirement***:** **IV:** Elemental magnesium 8 to 20 mEq (4 to 10 mmol) daily as component of parenteral nutrition; adjust daily dose based on serum magnesium and clinical considerations (Ref).

Torsades de pointes

Torsades de pointes (off-label use):

Polymorphic ventricular tachycardia (with pulse) associated with QT prolongation (torsades de pointes):

IV: 1 to 2 g (diluted in 50 to 100 mL D5W) over 15 minutes. If no response or torsades de pointes recurs, may repeat dose up to a total of 4 g in 1 hour; may follow with a continuous IV infusion of 0.5 to 1 g/hour (Ref).

Ventricular fibrillation/pulseless ventricular tachycardia associated with torsades de pointes:

Note: Administer in conjunction with electrical cardioversion/defibrillation**.**

IV, intraosseous: 2 g (diluted in 10 mL D5W) administered as a bolus over ≥1 to 2 minutes; if ineffective, may repeat immediately with 2 additional 2 g bolus doses as needed; maximum total dose: 6 g; use intraosseous route if IV not available (Ref).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: For indications that generally necessitate the use of ≤2 bolus doses (eg, treatment of asthma/chronic obstructive pulmonary disease exacerbations, torsades de pointes, symptomatic hypomagnesemia), no dosage adjustment is necessary for patients with kidney impairment or receiving kidney replacement therapies (Ref).

Altered kidney function:

General recommendations:

CrCl ≥30 mL/minute: **IV, IM:** No dosage adjustment necessary (Ref).

CrCl 10 to <30 mL/minute: **IV, IM:** Administer 50% of the usual indication-specific dose (Ref). Use with caution; frequent monitoring for hypermagnesemia is required.

CrCl <10 mL/minute: **IV, IM:** Hypermagnesemia develops frequently in patients with CrCl <10 mL/minute. If necessary, administer 50% of the usual indication-specific dose with extreme caution; frequent monitoring for hypermagnesemia is required (Ref).

Indication-specific recommendations:

Constipation:

CrCl ≥60 mL/minute: **Oral:** No dosage adjustment necessary; for occasional use only (Ref).

CrCl 30 to <60 mL/minute: **Oral:** Consider alternative agent. If used, administer doses on the low end of the recommended range and limit to occasional use only (Ref).

CrCl <30 mL/minute: Avoid use (Ref).

Eclampsia/Preeclampsia with severe features, seizure prophylaxis and treatment: An optimal regimen has not been identified; refer to institutional protocols. Close monitoring (including kidney function [eg, urine output], respiration, and patellar reflexes) is required; monitoring of magnesium levels is recommended for patients with preexisting or acute kidney impairment and in patients with signs of toxicity or seizure(s). Adjust dose to avoid maternal and fetal toxicity. Calcium gluconate should be available for treating severe magnesium toxicity.

Note: Patients with acute kidney injury (significant rise in creatinine above baseline) or oliguria (<30 mL/hour urine output for more than 4 hours) also may warrant dose reduction (Ref).

CrCl ≥60 mL/minute: **IV:** No dosage adjustment necessary (Ref).

CrCl 30 to <60 mL/minute: **IV:** Initial: 4 to 6 g loading dose over 15 to 30 minutes at onset of labor or induction/cesarean delivery, followed by 1 g/hour continuous infusion for at least 24 hours after delivery. **Note:** Monitor magnesium concentrations frequently (eg, at least every 4 hours) to avoid accumulation and adverse effects associated with hypermagnesemia (Ref).

CrCl <30 mL/minute: **IV:** Initial: 4 to 6 g loading dose over 15 to 30 minutes at onset of labor or induction/cesarean delivery. Continuous maintenance infusion is not recommended due to risk of accumulation, but intermittent magnesium supplementation may be administered as needed (Ref). **Note:** Use with extreme caution; monitor magnesium levels frequently (eg, at least every 2 to 4 hours) to inform the need for additional doses and avoid accumulation and adverse effects associated with hypermagnesemia (Ref). Per the manufacturer, do not exceed 20 g in patients with severe kidney impairment during a 48-hour period.

Fetal neuroprotection for imminent preterm birth (maternal administration) (off-label use): Note: An optimal regimen has not been identified; refer to institutional protocols. Close monitoring (including kidney function [eg, urine output], respiration, and patellar reflexes) is required (Ref).

Note: Patients with acute kidney injury (significant rise in creatinine above baseline) or oliguria (<30 mL/hour urine output for more than 4 hours) also may warrant dose reduction (Ref).

CrCl ≥60 mL/minute: **IV:** No dosage adjustment necessary (Ref).

CrCl 30 to <60 mL/minute: **IV:** Initial: 4 g loading dose over 20 to 30 minutes; may follow with a continuous infusion at 50% of usual infusion rate for a maximum of 24 hours or until delivery, whichever comes first. **Note:** Monitor magnesium levels frequently (eg, at least every 4 hours) to avoid accumulation and adverse effects associated with hypermagnesemia (Ref).

CrCl <30 mL/minute: **IV:** Initial: 4 g loading dose over 20 to 30 minutes. Continuous maintenance infusion is not recommended due to risk of accumulation, but intermittent magnesium supplementation may be administered if necessary (Ref). **Note:** Use with extreme caution; monitor magnesium levels frequently (eg, at least every 2 to 4 hours) to inform the need for additional doses and avoid accumulation and adverse effects associated with hypermagnesemia (Ref).

Hemodialysis, intermittent (thrice weekly): Dialyzable (extent unknown, removal will depend on magnesium concentration in dialysate) (Ref):

IV, IM:

General recommendations: Dose as for patients with CrCl <10 mL/minute (Ref).

Indication-specific recommendations: Dose as for patients with CrCl <30 mL/minute (Ref)

Oral: Avoid use (Ref).

Peritoneal dialysis:

IV, IM:

General recommendations: Dose as for patients with CrCl <10 mL/minute (Ref).

Indication-specific recommendations: Dose as for patients with CrCl <30 mL/minute (Ref).

Oral: Avoid use (Ref).

CRRT:

IV, IM: Dose as for patients with CrCl 10 to <30 mL/minute; consider magnesium concentration in dialysate and replacement fluids as well (Ref).

Oral: Avoid use (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration):

IV, IM: Dose as for patients with CrCl 10 to <30 mL/minute; consider magnesium concentration in dialysate and replacements fluids as well (Ref).

Oral: Avoid use (Ref).

Dosing: Liver Impairment: Adult

No dosage adjustment necessary.

Dosing: Obesity: Adult

Refer to indication-specific dosing for obesity-related information (may not be available for all indications).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see ”Magnesium sulfate: Pediatric drug information”)

Dosage guidance:

Dosing: 1,000 mg of magnesium sulfate = 98.6 mg **elemental** magnesium = 8.12 mEq **elemental** magnesium = 4.06 mmol **elemental** magnesium.

Clinical considerations: Serum magnesium is poor reflection of repletion status as the majority of magnesium is intracellular; serum concentrations may be transiently normal for a few hours after a dose is given, therefore, aim for consistently high normal serum concentrations in patients with normal renal function for most efficient repletion.

Hypomagnesemia

Hypomagnesemia:

Note: Dose depends on clinical condition and serum magnesium concentration.

Infants, Children, and Adolescents: Dose expressed as **magnesium sulfate:** IV, Intraosseous: 25 to 50 **mg**/kg/dose every 6 hours for 2 to 3 doses, then recheck serum concentration; maximum dose: 2,000 mg/dose (Ref).

Asthma, severe acute exacerbations

Asthma, severe acute exacerbations (adjunctive agent): Limited data available:

Emergency/intensive care management: Limited data available: **Note:** For severe, life-threatening exacerbations nonresponsive to initial therapy; not recommended for routine use (Ref). Dose expressed as **magnesium sulfate:**

Infants, Children, and Adolescents: IV: 50 mg/kg/dose as a single dose (Ref); usual dose range: 25 to 75 mg/kg/dose; maximum dose: 2,000 mg/dose (Ref).

Constipation, occasional

Constipation, occasional:

Note: With OTC use, should not exceed recommended treatment duration (7 days) unless directed by health care provider.

Children 6 to <12 years: Oral: 1 to 2 level teaspoons of granules dissolved in 8 ounces of water; may repeat in 4 hours. Do not exceed 2 doses per day.

Children ≥12 years and Adolescents: Oral: 2 to 4 level teaspoons of granules dissolved in 8 ounces of water; may repeat in 4 hours. Do not exceed 2 doses per day.

Parenteral nutrition, maintenance magnesium requirement

Parenteral nutrition, maintenance magnesium requirement:

Dose expressed as **elemental magnesium**:

Infants and Children ≤50 kg: IV: 0.3 to 0.5 **mEq**/kg/day as an additive to parenteral nutrition solution (Ref).

Children >50 kg and Adolescents: IV: 10 to 30 **mEq/**day as an additive to parenteral nutrition solution (Ref).

Torsade de pointes or ventricular fibrillation/pulseless ventricular tachycardia associated with torsade de pointes

Torsade de pointes or ventricular fibrillation/pulseless ventricular tachycardia associated with torsade de pointes: Infants, Children, and Adolescents: Dose expressed as **magnesium sulfate:** IV, Intraosseous: 25 to 50 **mg**/kg/dose; maximum dose: 2,000 mg/dose (Ref).

Dosing: Kidney Impairment: Pediatric

Hypomagnesemia: There are no dosage adjustments provided in the manufacturer's labeling; use with caution; monitor closely for hypermagnesemia.

Dosing: Liver Impairment: Pediatric

No dosage adjustment necessary.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse effects on neuromuscular function may occur at lower concentrations in patients with neuromuscular disease (eg, myasthenia gravis).

Frequency not defined:

Cardiovascular: Flushing (IV; dose related), hypotension (IV; rate related), vasodilation (IV; rate related)

Endocrine & metabolic: Hypermagnesemia

Contraindications

Hypersensitivity to any component of the formulation; heart block (see **Note**); myocardial damage; IV use for preeclampsia/eclampsia during the 2 hours prior to delivery (see **Note**)

Note: Although the manufacturers' labeling for some IV formulations state use in preeclampsia/eclampsia during the 2 hours prior to (cesarean) delivery is contraindicated due to interaction with neuromuscular-blocking agents intraoperatively; stopping magnesium sulfate prior to cesarean delivery in these patients is not recommended and increases the risk of seizure. Instead, magnesium should be continued prior to and during the delivery (ACOG 2013). Additionally, the manufacturers' labeling for some IV formulations contraindicate the use of magnesium sulfate in the setting of heart block; however, the use of magnesium is appropriate in patients with serious conditions requiring magnesium therapy who either have mild degrees of heart block (eg, first degree) or more severe forms of heart block with a temporary or permanent cardiac pacemaker.

Warnings/Precautions

Disease-related concerns:

• Neuromuscular disease: Use with extreme caution in patients with myasthenia gravis or other neuromuscular disease.

• Renal impairment: Use with caution in patients with renal impairment; accumulation of magnesium may lead to magnesium intoxication.

Special populations:

• Obstetrics: Vigilant monitoring and safe administration techniques (ISMP 2005) are recommended to avoid potential for errors resulting in toxicity. Monitor mother and fetus closely. Use longer than 5 to 7 days may cause adverse fetal events.

Dosage form specific issues:

• Aluminum: The parenteral product may contain aluminum; toxic aluminum concentrations may be seen with high doses, prolonged use, or renal dysfunction. Premature neonates are at higher risk due to immature renal function and aluminum intake from other parenteral sources. Parenteral aluminum exposure of >4 to 5 mcg/kg/day is associated with CNS and bone toxicity; tissue loading may occur at lower doses (Federal Register 2002). See manufacturer's labeling.

Other warnings/precautions:

• Appropriate use: Unlikely to effectively terminate irregular/polymorphic VT (with normal baseline QT interval) (AHA [Wigginton 2025]).

• Electrolyte abnormalities: Concurrent hypokalemia or hypocalcemia can accompany a magnesium deficit. Hypomagnesemia is frequently associated with hypokalemia and requires correction in order to normalize potassium.

• Parenteral administration: Magnesium toxicity can lead to fatal cardiovascular arrest and/or respiratory paralysis.

• Self-medication (OTC use): When used as a laxative, patients should consult a health care provider prior to use if they have: kidney disease; are on a magnesium-restricted diet; have abdominal pain, nausea, or vomiting; change in bowel habits lasting >2 weeks; have already used a laxative for >1 week.

Warnings: Additional Pediatric Considerations

Multiple salt forms of magnesium exist; close attention must be paid to the salt form when ordering and administering magnesium; incorrect selection or substitution of one salt for another without proper dosage adjustment may result in serious over- or underdosing.

Dosage Forms Considerations

1 g of magnesium sulfate = 98.6 mg of elemental magnesium = 8.12 mEq of elemental magnesium = 4.06 mmol of elemental magnesium.

Magnesium sulfate 1% [10 mg/mL] in Dextrose 5% injection is equivalent to elemental magnesium 0.081 mEq/mL.

Magnesium sulfate 2% [20 mg/mL] in Dextrose 5% injection is equivalent to elemental magnesium 0.162 mEq/mL.

Magnesium sulfate 4% [40 mg/mL] in Water injection is equivalent to elemental magnesium 0.325 mEq/mL.

Magnesium sulfate 8% [80 mg/mL] in Water injection is equivalent to elemental magnesium 0.65 mEq/mL.

Magnesium sulfate 50% injection is equivalent to elemental magnesium 4 mEq/mL.

Magnesium sulfate USP is supplied as magnesium sulfate heptahydrate

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

M2 Magnesium: 100 mg [DSC]

Granules, Oral:

Epsom Salt: (454 g, 1810 g)

FT Epsom Salt: (454 g)

GoodSense Epsom Salt: (454 g, 1810 g)

Solution, Injection:

Generic: 50% (10 mL, 20 mL)

Solution, Injection [preservative free]:

Generic: 50% (2 mL, 10 mL, 20 mL, 50 mL)

Solution, Intravenous:

Generic: 4 g/100 mL (100 mL); 1 g/100 mL (100 mL); 2 g/50 mL (50 mL); 20 g/500 mL (500 mL); 4 g/50 mL (50 mL); 40 g/1000 mL (1000 mL)

Solution, Intravenous [preservative free]:

Generic: 3% (100 mL); 4 g/100 mL (100 mL); 1 g/100 mL (100 mL); 2 g/50 mL (50 mL); 20 g/500 mL (500 mL); 4 g/50 mL (50 mL); 40 g/1000 mL (1000 mL [DSC])

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Magnesium Sulfate in D5W Intravenous)

1GM/100ML 5% (per mL): $0.03 - $0.11

Solution (Magnesium Sulfate Injection)

50% (per mL): $0.72 - $1.55

Solution (Magnesium Sulfate Intravenous)

2 gm/50 mL (per mL): $0.09 - $0.39

3 g/100 mL (per mL): $0.04

4 g/100 mL (per mL): $0.04 - $0.11

4 gm/50 mL (per mL): $0.10 - $0.19

20 g/500 mL (per mL): $0.01 - $0.04

40GM/1000ML (per mL): $0.01 - $0.03

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 40 mg/mL (50 mL, 100 mL); 20% (10 mL); 49.3% (500 mL); 50% (10 mL, 50 mL)

Administration: Adult

Injection: May be administered IM, intraosseous (IO), or IV.

IM: Adults: 50% concentration.

Eclampsia/preeclampsia: May mix with 1 mL lidocaine 2% to reduce injection pain (Ref).

IV, intraosseous: Must be diluted to a ≤20% solution for IV infusion and may be administered IVP, IVPB, as a continuous IV infusion, or IO. When giving IV push, must dilute first and should generally not be given any faster than 150 mg/minute. In patients not in cardiac arrest, hypotension and asystole may occur with rapid administration. For patients in cardiac arrest (ie, ventricular fibrillation or pulses ventricular tachycardia associated with torsades de pointes), may administer IV bolus (diluted in 10 mL D5W) over ≥1 to 2 minutes (Ref). Refer to indication-specific infusion rates in dosing for detailed recommendations.

Oral: When used as a laxative, dissolve dose in 8 ounces of water prior to ingesting. Lemon juice may be added to the solution to improve the taste.

Preparation for Administration: Adult

IV: Dilute to ≤20% in a compatible solution (eg, D5W, NS) for IV infusion.

IM: A 50% concentration may be used.

Oral: Dissolve granules in 8 ounces of water prior to administration. May add lemon juice to improve taste.

Administration: Pediatric

Oral: Must dissolve granules prior to administration. When used as a laxative, the patient should drink a full 8 ounces of liquid following each dose. Lemon juice may be added to the initial solution to improve the taste. **Note:** Most effective when taken on an empty stomach.

Parenteral:

IM: Infants and Children: Dilute prior to injection; in adults, doses are administered undiluted (50%) or diluted in a compatible fluid.

IV: Dilute in an appropriate fluid prior to administration. Rate of infusion dependent upon use:

Hypomagnesemia: **Note:** Rate should be slowed if patient experiences diaphoresis, flushing, or a warm sensation (Ref).

Moderate to severe hypomagnesemia: 0.1 to 0.25 **mEq**/kg/hour **elemental magnesium** (10 to 25 **mg**/kg/hour **magnesium sulfate**); usual maximum infusion rate in adults is 1,000 **mg**/hour of **magnesium sulfate** (Ref).

Urgent/emergent: Infuse slowly over 15 to 20 minutes (Ref).

Pulseless ventricular tachycardia (VT) associated with torsades: May administer as a bolus over several minutes (Ref).

VT with pulses associated with torsades: Infuse over 10 to 20 minutes; rapid infusion may cause hypotension (Ref).

Asthma exacerbation (severe): Infuse over 10 to 60 minutes (Ref).

Continuous IV infusion: After dilution, administer via an infusion pump.

Preparation for Administration: Pediatric

Oral: Dissolve granules in 8 ounces of water prior to administration.

Parenteral:

IM: Infants and Children: Dilute to a maximum concentration of 20% (ie, 200 mg/mL of **magnesium sulfate** or 1.6 mEq/mL of **elemental magnesium**) prior to administration.

IV: Dilute in a compatible solution (eg, D5W, NS) to a usual concentration of 0.5 mEq/mL of **elemental magnesium** (ie, 60 mg/mL of **magnesium sulfate**); maximum concentration: 20% (ie, 200 mg/mL of **magnesium sulfate** or 1.6 mEq/mL of **elemental magnesium**).

Storage/Stability

Prior to use, store at room temperature of 20°C to 25°C (68°F to 77°F). Do not freeze. Refrigeration of solution may result in precipitation or crystallization.

Use: Labeled Indications

Oral: Laxative for the relief of occasional constipation (OTC labeling).

Parenteral: Treatment and prevention of hypomagnesemia; prevention and treatment of seizures in eclampsia or preeclampsia with severe features; and treatment of cardiac arrhythmias (ventricular tachycardia/ventricular fibrillation) caused by hypomagnesemia.

Use: Off-Label: Adult

Asthma, severe acute exacerbation (adjunctive agent); Chronic obstructive pulmonary disease, severe acute exacerbation (adjunctive agent); Fetal neuroprotection for imminent preterm birth (maternal administration); Torsades de pointes

Medication Safety Issues

Sound-alike/look-alike issues:

Magnesium sulfate may be confused with manganese sulfate, morphine sulfate

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication (injectable formulation) among its list of drugs which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Other safety concerns:

$MgSO_4$ is an error-prone abbreviation (mistaken as morphine sulfate)

Metabolism/Transport Effects

None known.

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alfacalcidol: May increase serum concentrations of Magnesium Salts. Management: Consider using a non-magnesium-containing antacid or phosphate-binding product in patients also receiving alfacalcidol. If magnesium-containing products must be used with alfacalcidol, serum magnesium concentrations should be monitored closely. *Risk D: Consider Therapy Modification*

Alpha-Lipoic Acid: Magnesium Salts may decrease absorption of Alpha-Lipoic Acid. Alpha-Lipoic Acid may decrease absorption of Magnesium Salts. Management: Separate administration of alpha-lipoic acid from that of any magnesium-containing compounds by several hours. If alpha-lipoic acid is given 30 minutes before breakfast, then administer oral magnesium-containing products at lunch or dinner. *Risk D: Consider Therapy Modification*

Baloxavir Marboxil: Polyvalent Cation Containing Products may decrease serum concentrations of Baloxavir Marboxil. *Risk X: Avoid*

Bictegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Bictegravir. Management: Administer bictegravir at least 2 hours before or 6 hours after polyvalent cation containing products. Coadministration of bictegravir with or 2 hours after most polyvalent cation products is not recommended. *Risk D: Consider Therapy Modification*

Bisphosphonate Derivatives: Polyvalent Cation Containing Products may decrease serum concentrations of Bisphosphonate Derivatives. Management: Avoid administration of oral medications containing polyvalent cations within: 2 hours before or after tiludronate/clodronate/etidronate; 60 minutes after oral ibandronate; or 30 minutes after alendronate/risedronate. *Risk D: Consider Therapy Modification*

Cabotegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Cabotegravir. Management: Administer polyvalent cation containing products at least 2 hours before or 4 hours after oral cabotegravir. *Risk D: Consider Therapy Modification*

Calcitriol (Systemic): May increase serum concentrations of Magnesium Salts. Management: Consider using a non-magnesium-containing antacid or phosphate-binding product in patients also receiving calcitriol. If magnesium-containing products must be used, use caution and consider closely monitoring serum magnesium concentrations. *Risk D: Consider Therapy Modification*

Calcium Channel Blockers (Dihydropyridine): Magnesium Sulfate may increase adverse/toxic effects of Calcium Channel Blockers (Dihydropyridine). Specifically, the risk of hypotension or muscle weakness may be increased. *Risk C: Monitor*

Calcium Polystyrene Sulfonate: Laxatives (Magnesium Containing) may increase adverse/toxic effects of Calcium Polystyrene Sulfonate. More specifically, concomitant use of calcium polystyrene sulfonate with magnesium-containing laxatives may result in metabolic alkalosis or with sorbitol may result in intestinal necrosis. *Risk X: Avoid*

CNS Depressants: Magnesium Sulfate may increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Deferiprone: Polyvalent Cation Containing Products may decrease serum concentrations of Deferiprone. Management: Separate administration of deferiprone and oral medications or supplements that contain polyvalent cations by at least 4 hours. *Risk D: Consider Therapy Modification*

Dolutegravir: Magnesium Salts may decrease serum concentrations of Dolutegravir. Management: Administer dolutegravir at least 2 hours before or 6 hours after oral magnesium salts. Administer the dolutegravir/rilpivirine combination product at least 4 hours before or 6 hours after oral magnesium salts. *Risk D: Consider Therapy Modification*

Doxercalciferol: May increase hypermagnesemic effects of Magnesium Salts. *Risk C: Monitor*

Eltrombopag: Polyvalent Cation Containing Products may decrease serum concentrations of Eltrombopag. Management: Administer eltrombopag at least 2 hours before or 4 hours after oral administration of any polyvalent cation containing product. *Risk D: Consider Therapy Modification*

Elvitegravir: Polyvalent Cation Containing Products may decrease serum concentrations of Elvitegravir. Management: Administer elvitegravir-containing products 2 hours before, or 2 to 6 hours after, the administration of polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Gabapentin (Immediate Release): Magnesium Salts may increase CNS depressant effects of Gabapentin (Immediate Release). Specifically, high dose intravenous/epidural magnesium sulfate may enhance the CNS depressant effects of gabapentin. Magnesium Salts may decrease serum concentrations of Gabapentin (Immediate Release). Management: Administer gabapentin at least 2 hours after use of a magnesium-containing antacid. Monitor patients closely for evidence of reduced response to gabapentin therapy. Monitor for CNS depression if high dose IV/epidural magnesium sulfate is used. *Risk D: Consider Therapy Modification*

Levonadifloxacin: Magnesium Salts may decrease serum concentrations of Levonadifloxacin. *Risk X: Avoid*

Levothyroxine: Magnesium Salts may decrease serum concentrations of Levothyroxine. *Risk C: Monitor*

Multivitamins/Fluoride (with ADE): Magnesium Salts may decrease serum concentrations of Multivitamins/Fluoride (with ADE). Specifically, magnesium salts may decrease fluoride absorption. Management: To avoid this potential interaction separate the administration of magnesium salts from administration of a fluoride-containing product by at least 1 hour. *Risk D: Consider Therapy Modification*

Neuromuscular-Blocking Agents: Magnesium Salts may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

PenicillAMINE: Polyvalent Cation Containing Products may decrease serum concentrations of PenicillAMINE. Management: Separate the administration of penicillamine and oral polyvalent cation containing products by at least 1 hour. *Risk D: Consider Therapy Modification*

Phosphate Supplements: Magnesium Salts may decrease serum concentrations of Phosphate Supplements. Management: Administer oral phosphate supplements as far apart from the administration of an oral magnesium salt as possible to minimize the significance of this interaction. *Risk D: Consider Therapy Modification*

Quinolones: Magnesium Salts may decrease serum concentrations of Quinolones. Management: Administer oral quinolones several hours before (4 h for moxi/pe/spar/enox-, 2 h for others) or after (8 h for moxi-, 6 h for cipro/dela-, 4 h for lome/pe/enox-, 3 h for gemi-, and 2 h for levo-, nor-, or ofloxacin or nalidixic acid) oral magnesium salts. *Risk D: Consider Therapy Modification*

Raltegravir: Magnesium Salts may decrease serum concentrations of Raltegravir. Management: Avoid the use of oral / enteral magnesium salts with raltegravir. No dose separation schedule has been established that adequately reduces the magnitude of interaction. *Risk X: Avoid*

Ritodrine: May increase adverse/toxic effects of Magnesium Sulfate. *Risk C: Monitor*

Roxadustat: Polyvalent Cation Containing Products may decrease serum concentrations of Roxadustat. Management: Administer roxadustat at least 1 hour after the administration of oral polyvalent cation containing products. *Risk D: Consider Therapy Modification*

Sodium Polystyrene Sulfonate: Laxatives (Magnesium Containing) may increase adverse/toxic effects of Sodium Polystyrene Sulfonate. More specifically, concomitant use of sodium polystyrene sulfonate with magnesium-containing laxatives may result in metabolic alkalosis or with sorbitol may result in intestinal necrosis. *Risk X: Avoid*

Tetracyclines: Magnesium Salts may decrease absorption of Tetracyclines. Only applicable to oral preparations of each agent. Management: Avoid coadministration of oral magnesium salts and oral tetracyclines. If coadministration cannot be avoided, administer oral magnesium at least 2 hours before, or 4 hours after, oral tetracyclines. Monitor for decreased tetracycline therapeutic effects. *Risk D: Consider Therapy Modification*

Trientine: Polyvalent Cation Containing Products may decrease serum concentrations of Trientine. Management: Avoid concomitant use of trientine and polyvalent cations. If oral iron supplements are required, separate the administration by 2 hours. For other oral polyvalent cations, give trientine 1 hour before, or 1 to 2 hours after the polyvalent cation. *Risk D: Consider Therapy Modification*

Unithiol: May decrease therapeutic effects of Polyvalent Cation Containing Products. *Risk X: Avoid*

Food Interactions

Increased alcohol intake can deplete magnesium stores (IOM, 1997).

Pregnancy Considerations

Magnesium crosses the placenta; serum concentrations in the fetus are similar to those in the mother (Idama 1998; Osada 2002). Continuous maternal use for >5 to 7 days (in doses such as those used for preterm labor, an off-label use) may cause fetal hypocalcemia and bone abnormalities, as well as fractures in the neonate.

Magnesium sulfate injection is used for the prevention and treatment of seizures in pregnant or postpartum patients with eclampsia or preeclampsia with severe features. The risk of morbidity and mortality is increased in preeclampsia with severe features (ACOG 2020).

Magnesium sulfate may also be used prior to early preterm delivery for neuroprotection to reduce the risk of cerebral palsy (ACOG 2010; Reeves 2011); treatment may be of benefit when birth is anticipated before 32 weeks' gestation (ACOG 2016).

Tocolytics may be used for the short-term (48 hour) prolongation of pregnancy to allow for the administration of antenatal steroids and should not be used prior to fetal viability or when the risks of use to the fetus or mother are greater than the risk of preterm birth; maintenance therapy with tocolytics is ineffective and not recommended. Magnesium sulfate can be used up to 48 hours in patients at risk of delivery within 7 days; however, it is not the preferred tocolytic (ACOG 2016). Magnesium sulfate injection may be used in conjunction with other tocolytics for neuroprotection; however, an increased risk of maternal complications may be observed when used in combination with some tocolytic agents (ACOG 2016).

Breastfeeding Considerations

Magnesium is present in breast milk.

When magnesium sulfate is used in the intrapartum management of eclampsia, breast milk concentrations are generally increased for only ~24 hours after the end of treatment. In one study, this amounted to an increase of only 1.5 mg of magnesium to the breastfed infant on the first day after maternal therapy was stopped (Cruikshank 1982; Idama 1998).

Magnesium is endogenous to breast milk; concentrations remain constant during the first year of lactation and are not influenced by dietary intake under normal conditions (IOM 1997). Milk concentrations of magnesium are variable between lactating patients but are generally consistent within a given mother (Dórea 2000).

Magnesium requirements are the same in breastfeeding and nonbreastfeeding females (IOM 1997). Although the manufacturer recommends that caution be used if administered to lactating patients; magnesium sulfate when used for the prevention of seizures is considered compatible with breastfeeding (WHO 2002).

Dietary Considerations

Whole grains, legumes and dark-green leafy vegetables are dietary sources of magnesium (IOM 1997).

Adequate intake (AI) (elemental magnesium) (IOM 1997):

1 to 6 months: 30 mg daily

7 to 12 months: 75 mg daily

Dietary recommended daily allowance (RDA) (elemental magnesium) (IOM 1997):

1 to 3 years: 80 mg daily

4 to 8 years: 130 mg daily

9 to 13 years: 240 mg daily

14 to 18 years:

Females: 360 mg daily

Pregnancy: 400 mg daily

Lactation: 360 mg daily

Males: 410 mg daily

19 to 30 years:

Females: 310 mg daily

Pregnancy: 350 mg daily

Lactation: 310 mg daily

Males: 400 mg daily

≥31 years:

Females: 320 mg daily

Pregnancy: 360 mg daily

Lactation: 320 mg daily

Males: 420 mg daily

Monitoring Parameters

IV: Rapid administration: ECG monitoring, vital signs, deep tendon reflexes; magnesium concentrations if frequent or prolonged dosing required particularly in patients with renal dysfunction, calcium, and potassium concentrations; renal function.

Obstetrics: Patient status including vital signs, oxygen saturation, respiration, deep tendon reflexes, level of consciousness, fetal heart rate, maternal uterine activity, renal function. Monitor magnesium concentrations every 4 hours in patients with renal dysfunction (every 2 hours if serum magnesium is >8 mEq/L (ACOG 2020).

Reference Range

Note: Reference ranges may vary depending on the laboratory.

Magnesium serum concentration: Adults: 1.6 to 2.6 mg/dL (SI: 0.7 to 1.1 mmol/L or 1.3 to 2.1 mEq/L).

Eclampsia/preeclampsia therapeutic range: 5 to 9 mg/dL (SI: 2.1 to 3.7 mmol/L or 4.1 to 7.4 mEq/L) (ACOG 2020).

Mechanism of Action

When taken orally, magnesium promotes bowel evacuation by causing osmotic retention of fluid which distends the colon with increased peristaltic activity; parenterally, magnesium decreases acetylcholine in motor nerve terminals and acts on myocardium by slowing rate of S-A node impulse formation and prolonging conduction time. Magnesium is necessary for the movement of calcium, sodium, and potassium in and out of cells, as well as stabilizing excitable membranes.

Intravenous magnesium may improve pulmonary function in patients with asthma; causes relaxation of bronchial smooth muscle independent of serum magnesium concentration.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Antiseizure medication: IM: 1 hour; IV: Immediate; Laxative: Oral: 0.5 to 6 hours

Duration of antiseizure activity: IM: 3 to 4 hours; IV: 30 minutes

Absorption: Oral: Slow and poor (approximately one-third absorbed)

Distribution: Bone (50% to 60%); extracellular fluid (1% to 2%) (IOM 1997)

Protein binding: 30%, to albumin

Excretion: Urine (as magnesium); feces (as unabsorbed drug)

Patient Counseling Points

(For additional information see ”Magnesium sulfate: Patient drug information”)

What is this drug used for?

Injection:

• It is used to treat or prevent low magnesium levels.

• It is used to prevent and control seizures during pregnancy.

• It may be given to you for other reasons. Talk with the doctor.

Granules used by mouth:

• It is used to treat constipation.

Granules used as soaks:

• It is used to treat minor sprains or bruises.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Granules used by mouth:

• Diarrhea, upset stomach, or throwing up

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

• High magnesium levels like confusion, feeling sluggish, slow movements, shortness of breath, upset stomach, or very bad dizziness or passing out

• Low calcium levels like muscle cramps or spasms, numbness and tingling, or seizures

• Slow heartbeat

• Flushing

• Not able to move

• Change in eyesight

• Feeling cold

• Sweating a lot

• Muscle problem like new or worse muscle weakness, trouble chewing or swallowing, trouble breathing, droopy eyelids, or change in eyesight like blurred eyesight or seeing double

Granules used by mouth:

• Black, tarry, or bloody stools

• Cramps

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Sulfato de magnesio 25% inyectable lavimar | Sulfato de magnesio 25% norgreen | Sulfato de magnesio apolo;
- (AT) Austria: Cormagnesin | Magnesium sulfur;
- (AU) Australia: Epsom | Magnesium sulphate;
- (BD) Bangladesh: Nalepsin;
- (BE) Belgium: Magnesium sulfate sterop;
- (BG) Bulgaria: Cormagnesin | Magnesium sulfuricum;
- (BR) Brazil: Hypomagne | Magnoston | Sulfato de magnesio;
- (CO) Colombia: Sulfato de magnesio | Sulfato magnesia | Sulfato magnesio;
- (CR) Costa Rica: Sulfato de magnesio;
- (CZ) Czech Republic: Magnesium sulfuric;
- (DE) Germany: Lasar;
- (ES) Spain: Sulfato de magnesio genfarma;
- (FI) Finland: Addex-magnesium | Mg-konsentraatti;
- (FR) France: Spasmag | Sulfate de magnesium lavoisier;
- (GB) United Kingdom: Mag sulphate | Magnesium sulphate;
- (HU) Hungary: Cormagnesin;
- (ID) Indonesia: Magnesium sulphate;
- (IE) Ireland: Magnesium sulphate;
- (IN) India: Magneon;
- (IT) Italy: Magnesio solfato;
- (JP) Japan: Conclyte mg mitsubishi | Conclyte mg otsuka;
- (KE) Kenya: Nalepsin;

- (KW) Kuwait: Magnesium sulphate;
- (LT) Lithuania: Magnesii sulfas | Magnesium Sulfat | Magnesium sulfuric;
- (LV) Latvia: Magnesium sulfuric;
- (MA) Morocco: Spasmag;
- (MX) Mexico: Ifupeptol magnesiado | Sulfato de magnesio;
- (NO) Norway: Addex-magnesium | Magnesiumsulfat;
- (NZ) New Zealand: Magnesium sulphate;
- (PA) Panama: Sulfato de magnesio;
- (PE) Peru: Sulfagnesium;
- (PH) Philippines: Hizon magnesium sulfate;
- (PK) Pakistan: Magnesium sulphate;
- (PL) Poland: Magnesio solfato monico;
- (PY) Paraguay: Magnesio sulfato catedral;
- (RO) Romania: Cormagnesin;
- (RU) Russian Federation: Magnesium sulfate j | Magnesium sulfuric | Magnesium sulfuricum;
- (SE) Sweden: Addex-magnesium | Magnesio Solfato S.A.L.F.;
- (SG) Singapore: Magnesium sulphate | Magnesium Sulphate hospira;
- (SK) Slovakia: Magnesium sulfuricum;
- (TR) Turkey: Multiflex magnezyum sulfat | Pre eklamol magnezyum sulfat | Turktipsan magnezyum sulfat;
- (UA) Ukraine: Magnesium sulfuric;
- (UY) Uruguay: Sulfato de magnesio;
- (VE) Venezuela, Bolivarian Republic of: Sulfato de magnesio;
- (VN) Viet Nam: Magnesi bfs

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